STIVEN CATALYST

FIRST SERIES · OCTOBER 2026

MANAGEMENT REVIEW

Management without theatre.



AI

THE QUALITY OF DATA

How many new records carry an error, what bad data costs, six dimensions of quality, data cascades and wrong labels in AI, and a card for measuring your data in one afternoon.

47% of newly created records had at least one critical error, on average, in 75 checks that managers ran on their own units.

NAGLE, REDMAN & SAMMON, 2017

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Every report, KPI and AI model is only as good as the records behind it. How many of those records are wrong, what does that cost, and how can a team find out in one afternoon?

When 75 managers checked the newest records of their own units, 47% had at least one critical error on average, and fewer than 3% of the scores met the bar they set themselves. Estimates of the cost run from Gartner's $12.9 million a year per organisation to IBM's $3.1 trillion for the US, neither with a visible method. In AI, 92% of 53 practitioners interviewed had met a data cascade, and the test sets of widely used benchmarks hold at least 3.3% wrong labels on average.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

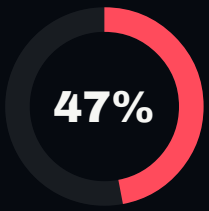
Our reading
Where the editor interprets rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

47 out of 100 new records

Over two years, 75 managers ran the same test on their own units: they took the last 100 records their teams had completed and counted those without a critical error, one that affects the work.



of newly created records had at least one critical error, on average.

Asked how good the data had to be, no manager called a score below the high nineties acceptable, and fewer than 3% of the scores reached that bar. The scores ranged from 0 to 99, and the authors found no meaningful differences between industries.

OUR READING

The test takes an afternoon and needs no software. The hard part is looking at the records your own team made.

Source: Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

75 measurements by managers in their own units: a small, unrepresentative sample. The bar was the managers' own, and the 3% is a share of measurements; companies were not counted.

THE NUMBERS

What bad data costs

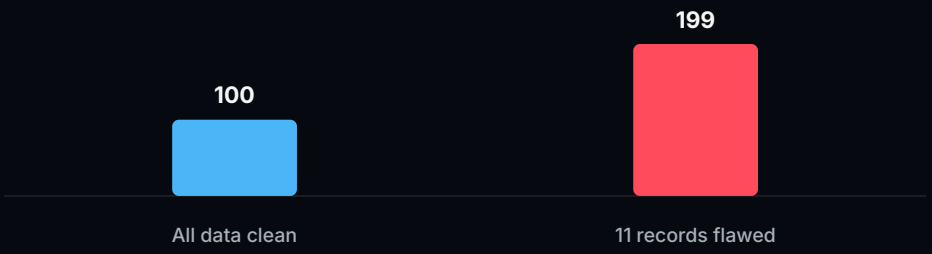
Nobody has measured the full cost of poor data. What exists are estimates and rules of thumb, worth reading together with their caveats.

$12.9m
a year: Gartner's average cost of poor data quality per organisation (2021)

50%
of knowledge workers' time lost in hidden data factories, by Redman's estimate

44%
of German firms that considered AI but do not use it cite the availability or quality of data (2025)

COST OF 100 UNITS OF WORK AT $1 EACH, BY REDMAN'S RULE OF TEN ($)



Redman calls the extra steps a team adds to cope with errors made upstream hidden data factories, and his rule of ten says flawed data make a unit of work cost ten times as much. Both are his estimates; neither was measured. The best-known figure, $3.1 trillion a year for the US in 2016, was IBM's; Redman later wrote that it could be off by a trillion dollars either way.

Sources: Manasi Sakpal, Gartner, 2021 (via Excerpts, through web search); Thomas C. Redman, Harvard Business Review, 2016 (via SAP Community, 2023); Statistisches Bundesamt (Destatis), 2025; Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

These are estimates, so do not add them up or compare them. Gartner does not show how it reached its average; the German figure is a reason firms report and measures no data quality.

THE MODEL

Six dimensions of quality

In 2013 a working group of DAMA UK, the data management association, named six primary dimensions for assessing data quality. The UK government adopted them in its 2020 data quality framework.

01
Completeness
nothing that should be there is missing

02
Uniqueness
no record appears twice

03
Timeliness
changes at the source show up quickly where the data are used

04
Validity
values fit the agreed format and rules

05
Accuracy
the data reflect reality

06
Consistency
the data agree across sources, which does not make them accurate

For AI, Gartner sets a different test: data are AI-ready when they represent the use case, including the patterns, errors and outliers a model needs. In a 2024 Gartner survey of 1,203 data management leaders, 63% said their organisation did not have, or was not sure it had, the right data practices for AI.

OUR READING

Clean and fit for use are two different things. Always ask: good enough for what?

Sources: DAMA UK Working Group, 2013 (via A copy of the October 2013 final version); Gartner, 2025

The definitions condense the DAMA UK document; DAMA-DMBOK and other sources give longer lists. The 63% is a self-report.

More in the essay Mistakes get lost between departments

WHAT THE RESEARCH SAYS

Everyone wants the model work

Nithya Sambasivan and colleagues at Google interviewed 53 practitioners who build AI for high-stakes uses in India, East and West Africa and the US. 92% had met at least one data cascade: a data problem that compounds and harms results further down the line.

WRONG LABELS IN THE TEST SETS OF WIDELY USED BENCHMARKS (%)



Curtis Northcutt, Anish Athalye and Jonas Mueller found at least 3.3% wrong labels on average in the test sets of ten widely used datasets of images, text and audio. Such errors can reverse the ranking of models: on ImageNet, the smaller ResNet-18 beats ResNet-50 if the share of originally mislabelled test examples rises by just 6%.

OUR READING

A model is judged against data, so wrong data mean a wrong judgment.

Sources: Nithya Sambasivan et al., ACM CHI, 2021; Curtis G. Northcutt, Anish Athalye & Jonas Mueller, NeurIPS Datasets and Benchmarks, 2021

The interviews cover 53 people and are not a representative survey; the test sets are public and come from no company. Versions of the paper give 3.3% or 3.4%.

HOW IT IS MEASURED

The Friday afternoon measurement

The method of Nagle, Redman and Sammon needs no software: a team, an afternoon and the last 100 records it completed.

1. **Take the last 100 records**
   the ones your unit completed most recently
2. **Pick the 10–15 most important fields**
   the ones the next step depends on
3. **Mark every visible error**
   record by record, with people who know the data
4. **Count the error-free records**
   that number, from 0 to 100, is the score

HYPOTHETICAL EXAMPLE, 100 CUSTOMER ORDERS

Errors	wrong address in 9, no phone number in 14, wrong quantity in 5; 4 orders have two errors
Score	24 orders have an error, 76 are clean: score 76
Rule of ten	$100 of work costs 76 + 24 × 10 = $316

Orders and errors alike are made up.

Source: Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

The steps follow Nagle, Redman & Sammon (2017); the rule of ten is Redman's rule of thumb; the example is the editors'.

TOOL OF THE ISSUE

The data quality card

One card per check, once a month. Count the errors by field, then sort them: a few fields usually carry most of them.

01 Records checked which 100, from which step, on which date

02 Critical fields the 10–15 fields the next step depends on

03 Errors by field missing, wrong, duplicate or out of date

04 Score error-free records out of 100

05 Where errors start the step or system where most are created

06 Fix at the source one change, who, by when; measure again next month

Sources: Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017; DAMA UK Working Group, 2013 (via A copy of the October 2013 final version)

Suggested by the editors, after the Friday afternoon measurement (2017) and the DAMA UK dimensions (2013).

Open the tool: Pareto 80/20 stivencatalyst.com/tools/pareto/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Only 3% of Companies' Data Meets Basic Quality Standards
Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017
https://hbr.org/2017/09/only-3-of-companies-data-meets-basic-quality-standards

Gründe gegen die Nutzung von Technologien der künstlichen Intelligenz nach Beschäftigtengrößenklassen
Statistisches Bundesamt (Destatis), 2025
https://www.destatis.de/DE/Themen/Branchen-Unternehmen/Unternehmen/IKT-in-Unternehmen-IKT-Branche/Tabellen/ikti-gegen-nutzung-kuenstliche-intelligenz.html

"Everyone wants to do the model work, not the data work": Data Cascades in High-Stakes AI
Nithya Sambasivan et al., ACM CHI, 2021
https://research.google/pubs/everyone-wants-to-do-the-model-work-not-the-data-work-data-cascades-in-high-stakes-ai/

Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks
Curtis G. Northcutt, Anish Athalye & Jonas Mueller, NeurIPS Datasets and Benchmarks, 2021
https://arxiv.org/abs/2103.14749

Bad Data Costs the U.S. $3 Trillion Per Year
Thomas C. Redman, Harvard Business Review, 2016 · via SAP Community, 2023
https://hbr.org/2016/09/bad-data-costs-the-u-s-3-trillion-per-year

Lack of AI-Ready Data Puts AI Projects at Risk
Gartner, 2025
https://www.gartner.com/en/newsroom/press-releases/2025-02-26-lack-of-ai-ready-data-puts-ai-projects-at-risk

How to Improve Your Data Quality
Manasi Sakpal, Gartner, 2021 · via Excerpts, through web search
https://www.gartner.com/smarterwithgartner/how-to-improve-your-data-quality

The Six Primary Dimensions for Data Quality Assessment
DAMA UK Working Group, 2013 · via A copy of the October 2013 final version

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked "via". A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked "Our reading".



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html

Management Review · No. 30 · October 2026 · Stiven Catalyst
Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html